

FUEL CELL MONITORING SYSTEM

Six-Layer PCB Design & Hardware Engineering Report

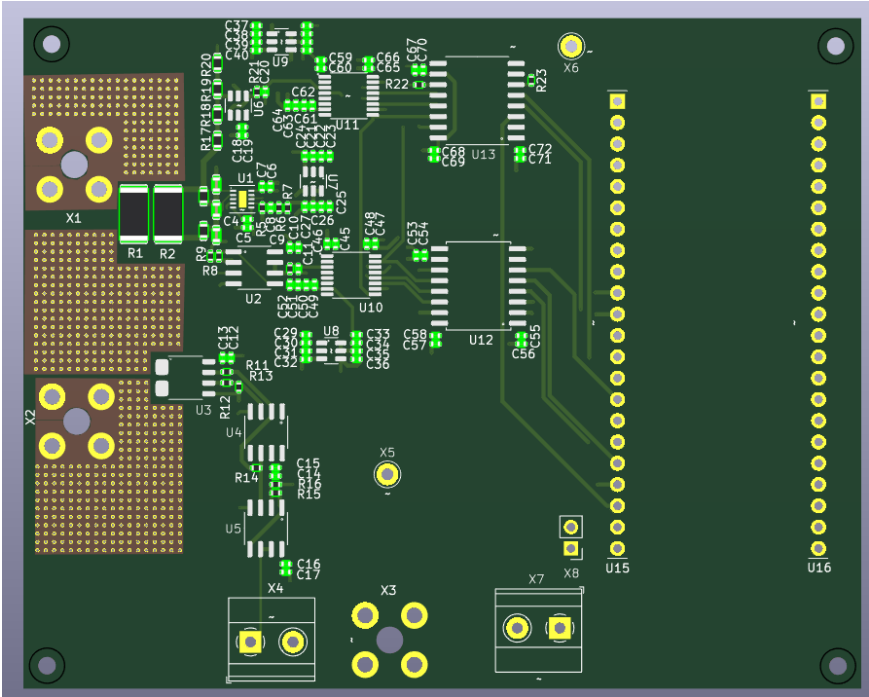


Figure 1. 3D visualization of the completed six-layer PCB layout.

Project	Fuel Cell Monitoring System
PCB Technology	6-layer printed circuit board
EDA Tool	KiCad
Controller	ESP32-S3
Measurement Functions	Fuel-cell voltage and current monitoring
Author	Samir Patel
Institution	Hochschule Wismar – Faculty of Engineering Sciences

Engineering Report

This report documents the schematic architecture, measurement signal conditioning, ADC and isolation interfaces, reference-voltage generation, microcontroller interface, and six-layer PCB implementation developed for a fuel-cell monitoring application.

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1. Project Overview

The project is a dedicated electronic monitoring board for a fuel-cell system. The hardware is designed to acquire electrical operating quantities, condition them for accurate analog-to-digital conversion, provide isolation where required, and transfer the resulting measurement data to an ESP32-S3 microcontroller through digital interfaces.

Main engineering objectives

- Measure the fuel-cell output voltage using a conditioned analog measurement path.
- Measure fuel-cell current using a dedicated current-sensing stage.
- Scale, buffer and filter analog signals before ADC conversion.
- Provide stable analog reference voltages for repeatable measurement.
- Use digital isolation between measurement-side circuitry and the controller-side interface.
- Implement the complete design as a compact six-layer PCB in KiCad.
- Provide a layout suitable for further prototype fabrication, assembly and laboratory validation.

Design philosophy

The design separates the analog measurement functions from the digital controller section. This is important because low-level measurement signals are sensitive to supply noise, return-current paths and digital switching activity. The use of dedicated filtering, buffering, reference circuitry and isolation supports a more controlled measurement environment.

2. System Architecture

The schematic is organized into functional blocks rather than treating the PCB as a single circuit. The principal signal flow is: fuel-cell input → sensing → analog conditioning → ADC → digital isolation → ESP32-S3 controller. Separate reference and supply-conditioning circuits support the measurement stages.

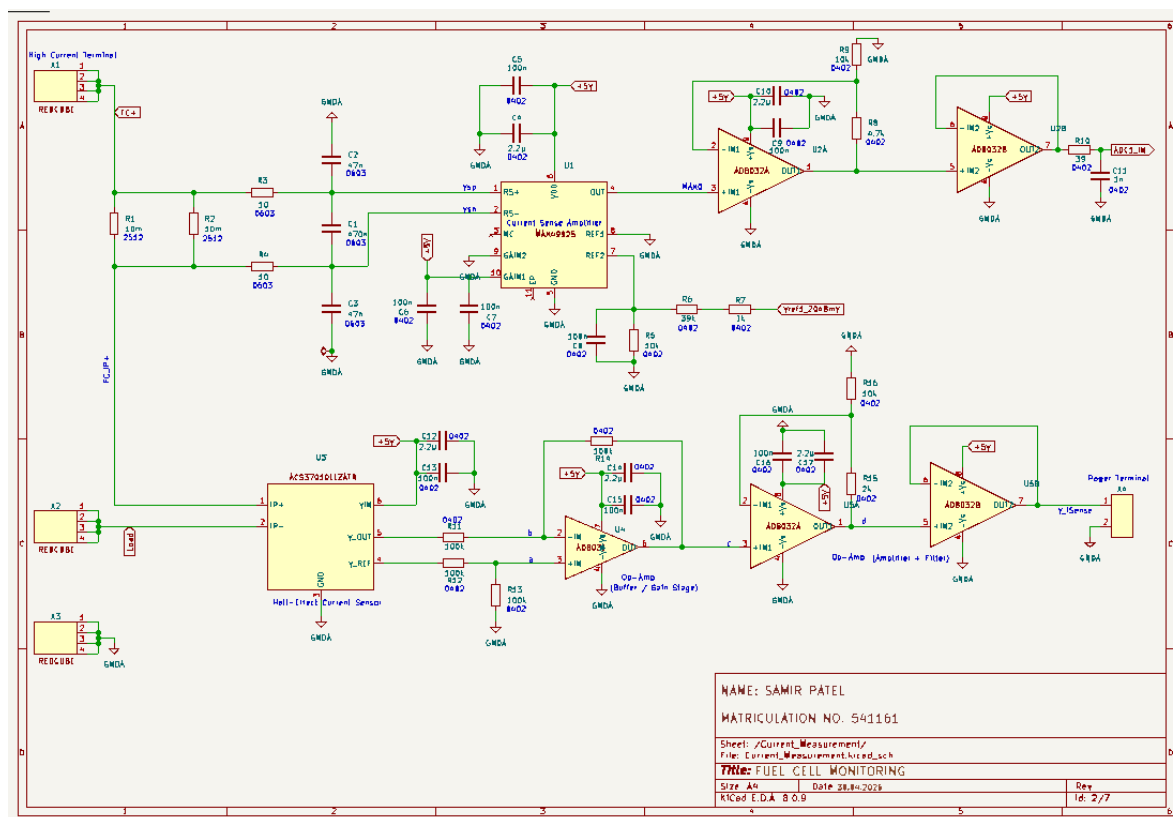


Figure 3. Main measurement circuitry showing voltage/current sensing, analog conditioning and output stages.

Functional blocks visible in the design

- Fuel-cell electrical input and high-current connection interfaces.
- Voltage sensing network with resistor-divider scaling and filtering.
- Current measurement using a dedicated current-sensing device and amplifier chain.
- Operational-amplifier stages for buffering, gain and signal conditioning.
- Precision voltage-reference sections with local decoupling.
- High-resolution ADC interfaces for digitizing conditioned measurement signals.
- Digital isolators for separation of the measurement and controller domains.
- ESP32-S3 interface and SPI/control connections.

3. Voltage Measurement Chain

The voltage-monitoring section converts the fuel-cell voltage into a controlled low-voltage analog signal suitable for the ADC input range. The front end uses resistor scaling, followed by filtering and an amplifier/buffer stage. This approach prevents the high-voltage source from being connected directly to the ADC while maintaining a defined measurement relationship.

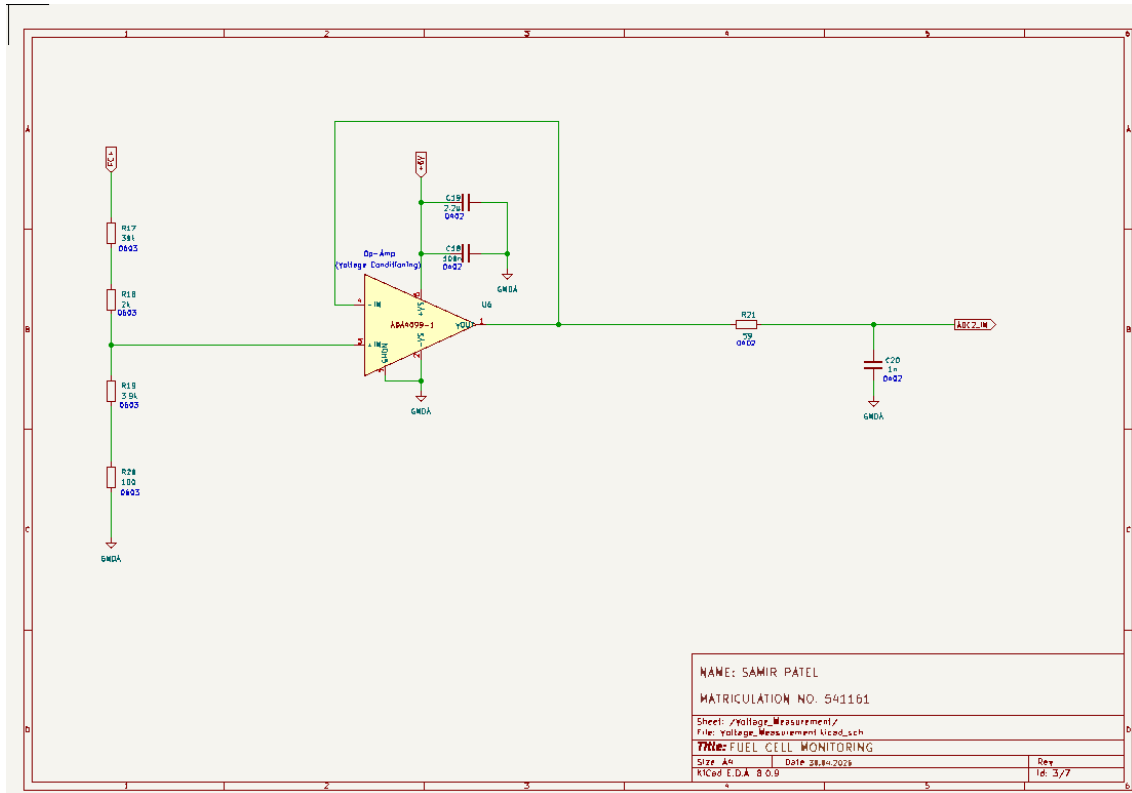


Figure 4. Voltage measurement and conditioning stage in KiCad.

Key design considerations

- The resistor divider establishes the input scaling ratio.
- RC filtering reduces high-frequency noise before conversion.
- The op-amp stage provides buffering and/or additional signal conditioning.
- The final signal is routed to the ADC through a controlled analog path.
- Local supply decoupling is used around active devices.

Measurement relationship

For a basic divider, the scaled voltage is determined by the resistor ratio. During laboratory calibration, the actual board should be characterized using a known reference voltage and the measured ADC code should be mapped to the physical fuel-cell voltage. Component tolerances, amplifier offset and ADC reference accuracy should be included in the final error budget.

4. Current Measurement Chain

The current-monitoring path uses a dedicated current-sensing element followed by analog signal conditioning. The schematic includes amplifier stages that prepare the sensor output for ADC conversion. The design therefore allows the low-level current-sense signal to be amplified and filtered before digitization.

Signal-conditioning strategy

- Current sensor generates a signal proportional to the measured current.
- An amplifier stage increases the useful signal level.
- RC networks provide bandwidth limitation and noise suppression.
- The conditioned signal is transferred to the ADC input.
- The final current value can be calculated in firmware from ADC code and the calibrated sensor transfer function.

The current channel should be calibrated at several known operating points rather than using only one point. A two-point or multi-point calibration can compensate for gain and offset errors and provide better accuracy over the operating range.

5. Voltage References and Power Integrity

Stable reference voltages are essential for accurate ADC measurements. The design contains dedicated reference-voltage circuits with multiple bypass capacitors placed around the reference devices. The repeated reference blocks indicate an intentional effort to provide clean reference rails for the measurement channels.

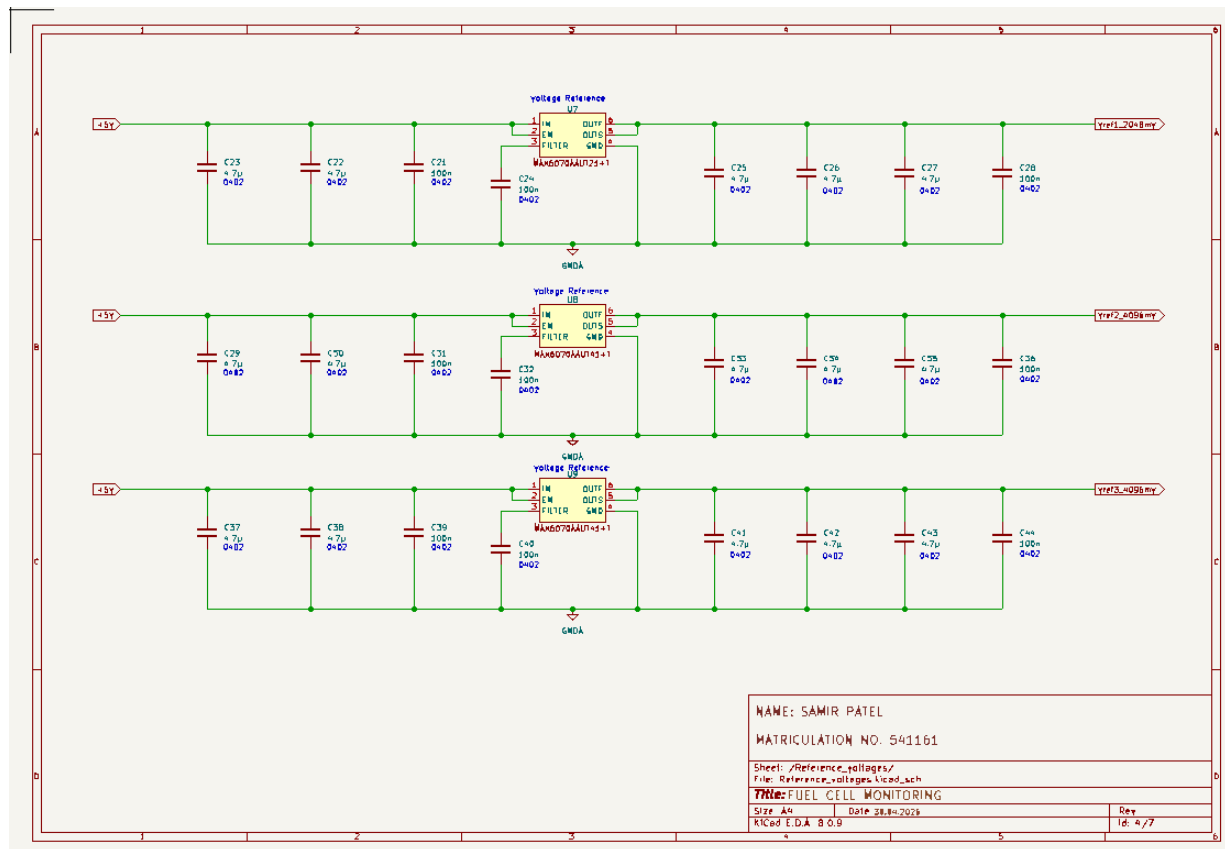


Figure 5. Dedicated voltage-reference circuits with local bypass capacitors.

Power-integrity practices

- Place high-frequency bypass capacitors close to IC supply/reference pins.
- Keep sensitive reference routing away from noisy digital traces where practical.
- Use a controlled return path for analog circuitry.
- Avoid unnecessary shared impedance between high-current and low-level measurement returns.
- Verify reference output voltage and noise experimentally during board bring-up.

6. ADC, Digital Isolation and SPI Interface

The measurement signals are digitized by ADC circuitry and then transferred across digital isolation to the controller domain. The schematics show dedicated digital isolators between the ADC-side signals and the microcontroller-side SPI/control signals. This architecture reduces the direct coupling of the measurement domain to the digital controller domain.

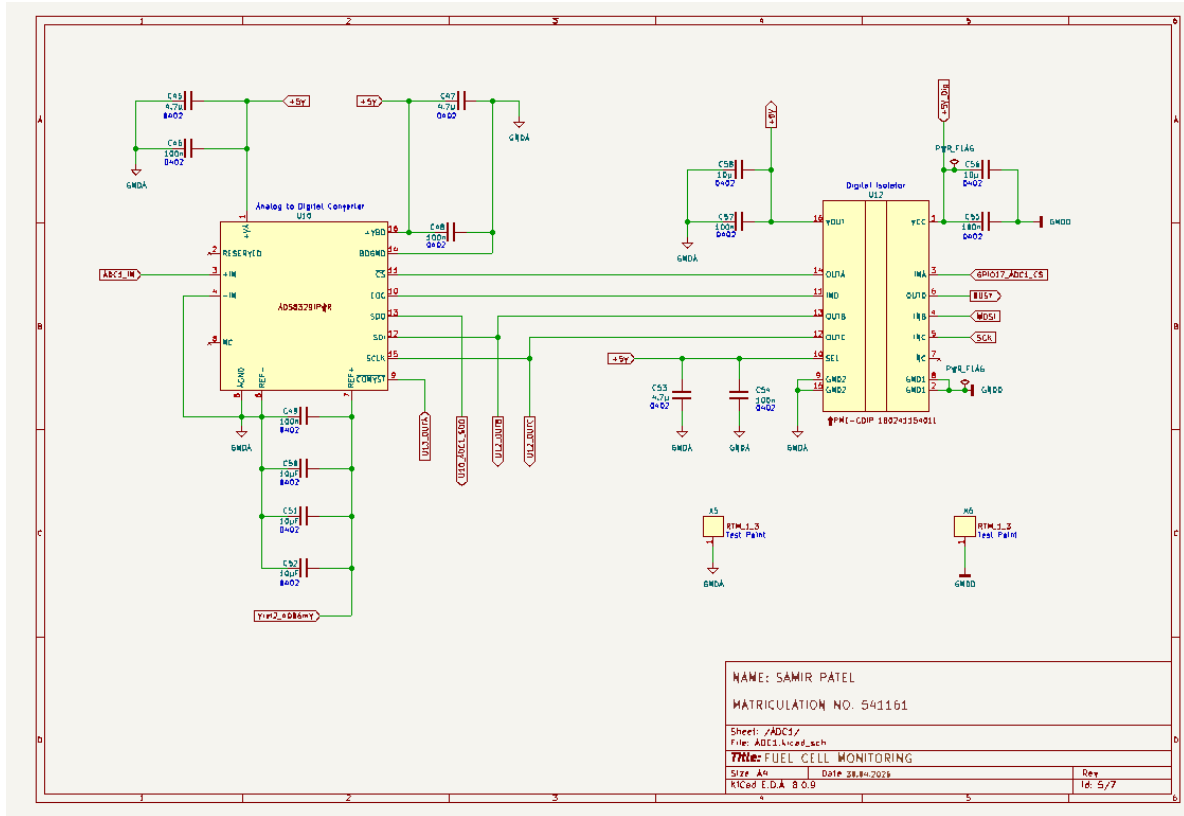
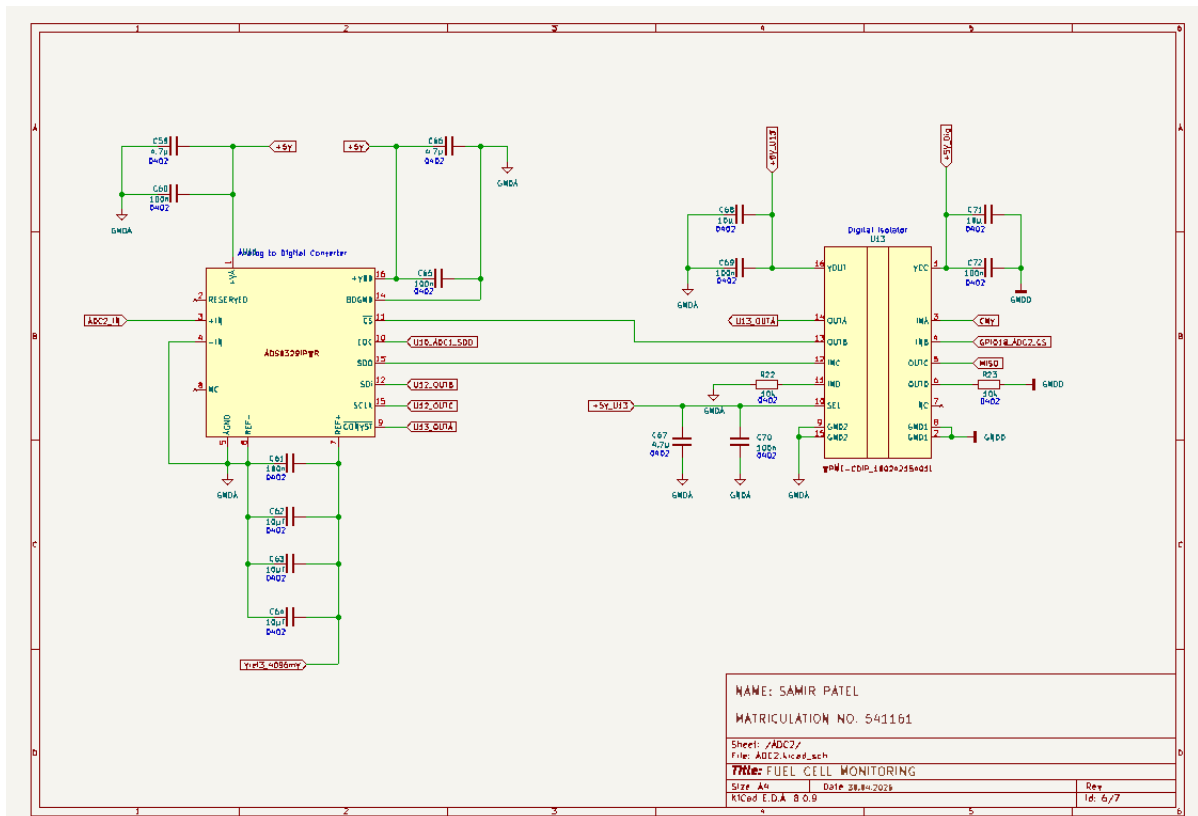


Figure 6. ADC and digital-isolation interface – measurement channel 1.



Interface signals

- SPI clock and data signals connect the ADC and controller interface.
- Chip-select and control lines are routed through the digital interface.
- Isolated power domains are supported by local decoupling.
- The PCB layout keeps the digital interface physically separated from the most sensitive analog nodes where practical.

7. Microcontroller Interface

The controller section is centered around an ESP32-S3 module. The schematic provides the required power, ground and digital signal connections for communication with the measurement subsystem. The large board-side header area provides flexibility for programming, debugging and system integration.

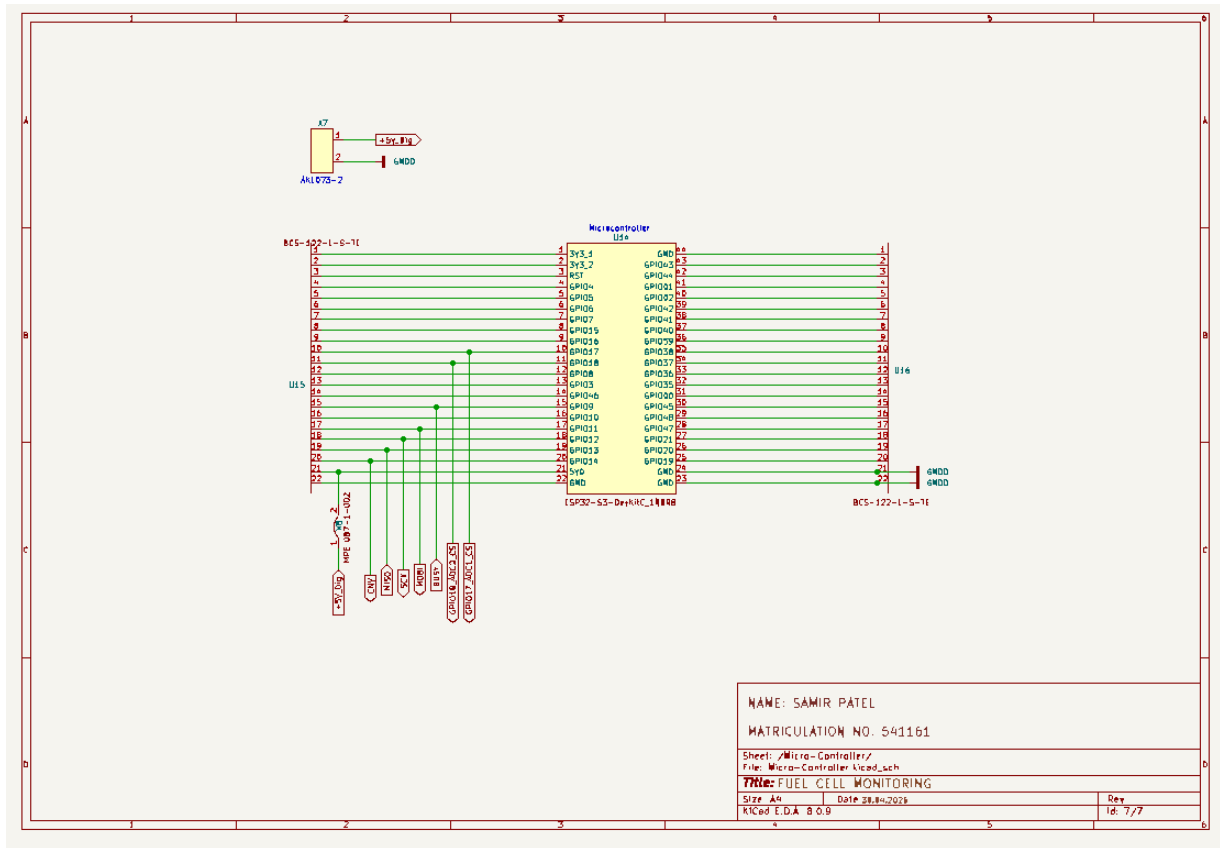


Figure 8. ESP32-S3 microcontroller and external interface connections.

Recommended firmware responsibilities

- Initialize the ADC/SPI interface and digital-isolation control signals.
- Read raw ADC conversion results.
- Apply gain, offset and calibration coefficients.
- Convert ADC results into engineering units such as volts and amperes.
- Provide diagnostic checks for out-of-range or invalid measurements.
- Transmit measurements through the selected communication interface.

8. Six-Layer PCB Design and Layout

The board was implemented as a six-layer PCB. The higher layer count provides additional routing and plane resources compared with a basic two-layer design and is particularly useful for a mixed-signal board containing precision analog circuitry, ADCs, digital isolation and a microcontroller interface.

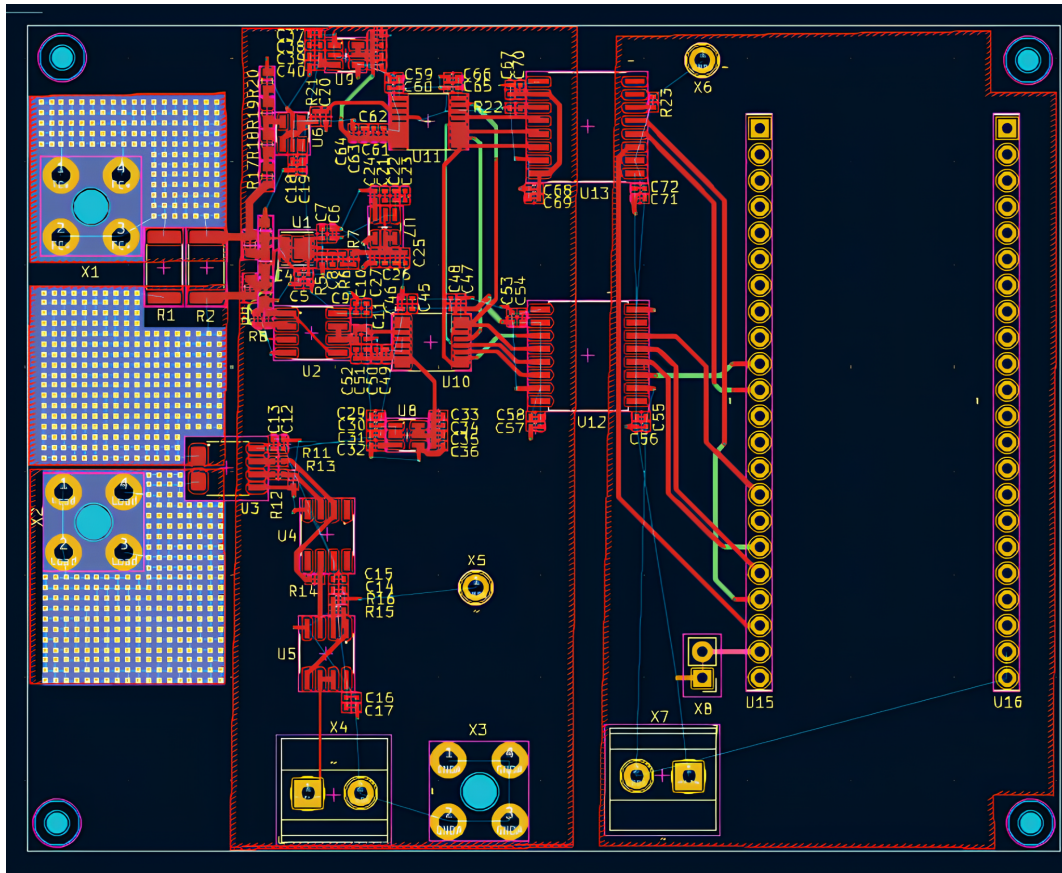


Figure 2. KiCad PCB layout view showing component placement and routed traces.

Layout strategy

- High-current/input connector regions are kept toward the board edge.
- Analog sensing and conditioning circuitry is concentrated in the left/central area.
- ADC and isolation circuitry forms an interface between analog and digital sections.
- The controller/header area is separated from the most sensitive analog circuitry.
- Large connector/header footprints provide system-level access and expansion.
- Mounting holes and mechanical keep-out areas are incorporated into the board outline.

Six-layer mixed-signal considerations

- Use dedicated or carefully controlled ground/reference planes.
- Keep fast digital routes away from sensitive analog inputs.
- Minimize loop area for ADC reference, amplifier feedback and decoupling paths.
- Route high-current paths with appropriate copper width and thermal considerations.
- Use short connections between IC supply pins and local bypass capacitors.
- Validate all clearances, drill sizes, annular rings and copper-to-edge rules against the selected PCB manufacturer.

9. Design-for-Manufacture and Verification

Before fabrication, the KiCad project should be treated as a manufacturing data package rather than only a layout. The following checks are recommended for the final revision.

Area	Verification item	Status
Schematic	ERC completed and all intentional warnings reviewed	Check before release
PCB	DRC completed with no unexplained violations	Check before release
Footprints	Pad geometry, pin-1 orientation and mechanical dimensions verified	Check before release
Layers	Six copper layers and stack-up confirmed with manufacturer	Confirmed by project
Power	Supply rails and polarity checked	Check before release
References	Reference outputs measured under load	Prototype test
ADC	Input range and reference voltage verified	Prototype test
Isolation	Isolation barrier and creepage/clearance verified	Prototype test
Calibration	Voltage/current calibration coefficients established	Prototype test
Fabrication	Gerbers, drill files and assembly documentation generated	Release step

Recommended release package

- KiCad project files: schematic, PCB layout, symbols and footprints.
- Fabrication outputs: Gerber and drill files for the six-layer stack-up.
- Bill of materials (BOM) with manufacturer part numbers.
- Pick-and-place data if automated assembly is used.
- PDF schematic and PCB drawings for documentation.
- Calibration and test procedure for prototype validation.

10. Engineering Assessment

The design demonstrates a complete hardware-development workflow from electrical sensing and analog conditioning through data conversion, digital isolation and microcontroller interfacing. The use of a six-layer PCB is appropriate for the density and mixed-signal nature of the design and gives the layout additional resources for signal integrity and power/ground distribution.

Strengths of the current design

- Clear separation of major analog, conversion, isolation and controller functions.
- Dedicated analog conditioning before ADC conversion.
- Dedicated voltage-reference circuitry with multiple local capacitors.
- Digital isolation between measurement and controller interfaces.
- Compact six-layer PCB implementation with integrated connectors and mounting features.
- KiCad-based design is suitable for reproducible revision and manufacturing documentation.

Areas to improve in the next revision

- Perform a complete quantitative error budget for voltage and current channels.
- Document the exact six-layer stack-up, copper thickness and impedance requirements.
- Add clearly documented test points for each major analog signal and supply rail.
- Review analog/digital return-current paths with particular attention to ADC reference and isolation boundaries.
- Complete prototype measurements for noise, offset, gain error and temperature drift.
- Add revision-controlled manufacturing and assembly documentation.

11. Conclusion and Future Improvements

A complete six-layer PCB-based fuel-cell monitoring platform has been developed in KiCad. The board integrates fuel-cell voltage and current measurement, analog signal conditioning, precision reference circuitry, ADC conversion, digital isolation and ESP32-S3 interfacing on a single hardware platform.

The next stage should focus on prototype fabrication and laboratory characterization. In particular, the voltage and current channels should be calibrated against traceable instruments, while ADC noise, amplifier offset, reference stability and thermal behavior should be measured. These results can then be used to refine component values, PCB routing and firmware calibration.

Future development opportunities

- Add automated self-test and sensor fault detection.
- Implement digital filtering and configurable sampling rates in firmware.
- Add non-volatile storage for calibration coefficients.
- Provide a dedicated communication interface for PC or cloud-based monitoring.
- Improve test-point accessibility for production and laboratory debugging.
- Document and validate the complete PCB stack-up and EMC/EMI performance.

Document status	Professional project documentation
PCB revision	Six-layer design
Design tool	KiCad
Primary controller	ESP32-S3
Next milestone	Prototype fabrication and electrical validation

End of Report